

Lab #8

Interfacing the LCD to the PIC18F4321

ECE 3301L Introduction to Microcontrollers Laboratory

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Videos:

Prelab:

Working prelab code:

<https://youtu.be/XHqCQq1aTBE?si=ArAScL64RtX25TBY>

Working prelab circuit:

- <https://youtube.com/shorts/VwMIUFFOeiU?si=7lyhedz3OugZDae3>

Postlab:

Working postlab code:

https://youtu.be/DkiDRi_tTuc?si=TYZhz-H3TRKT3cdz

Working postlab circuit:

- <https://youtube.com/shorts/rBvM7w-zItA?si=goyPMpsdUc-kncSQ>
- <https://youtube.com/shorts/c7cyojLiqjs?si=VR6WFHhuaUNc5on5>

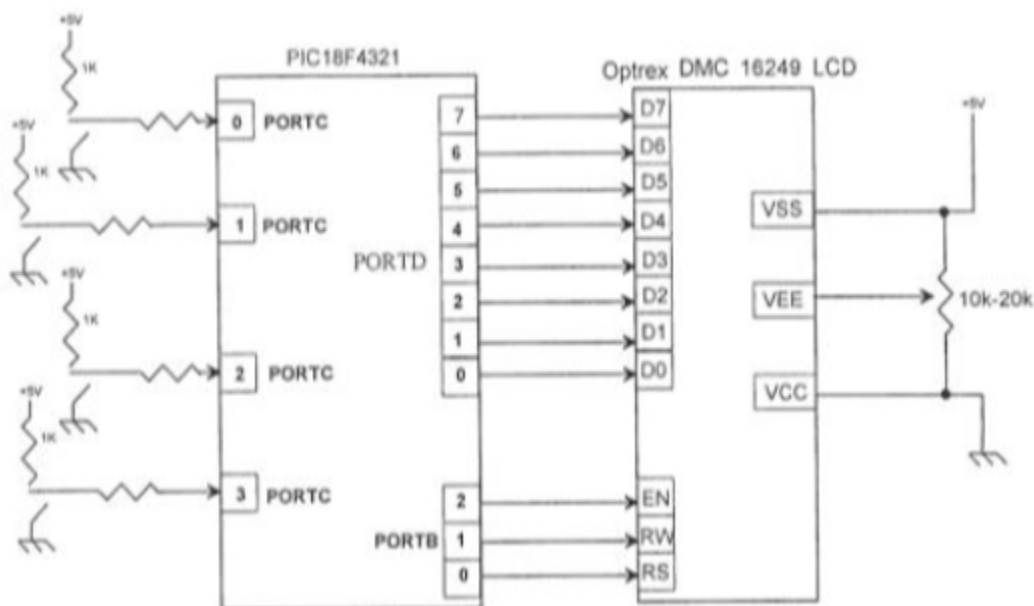
Objective:

The purpose of this lab is to interface an LCD display to the PIC18F4321 chip, and then write a PIC 18F assembly language program to display the message 'PIC18F'.

PRELAB:

prelab i)

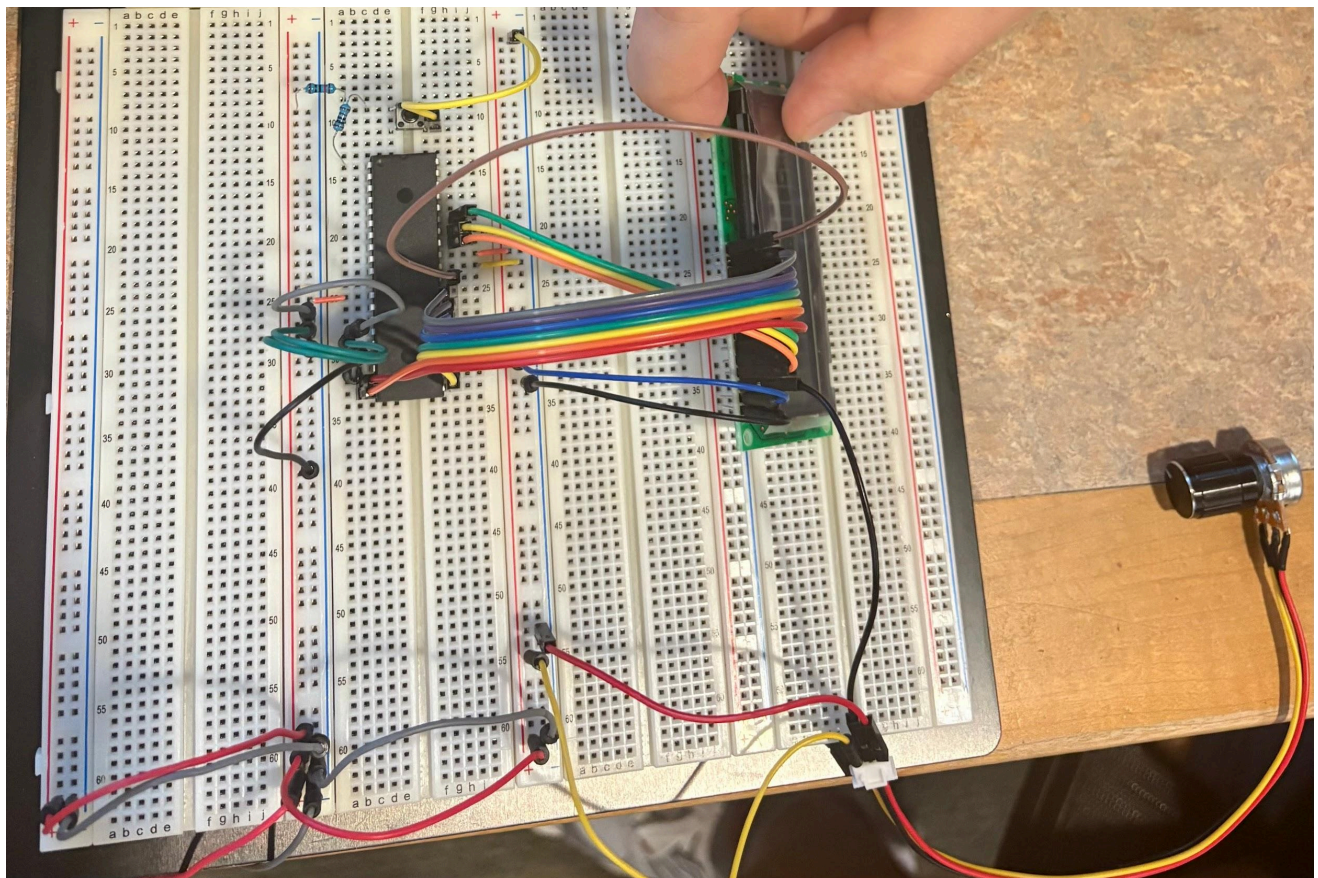
Assume the PIC18F4321- DMC 16249 interface of image 1. It is desired to display the phrase "PIC18F" on the LCD as soon as the four input switches connected to PORT C are all HIGH. Write a PIC18F assembly language program to accomplish this.



IMG 1. *The circuit build in the lab.*

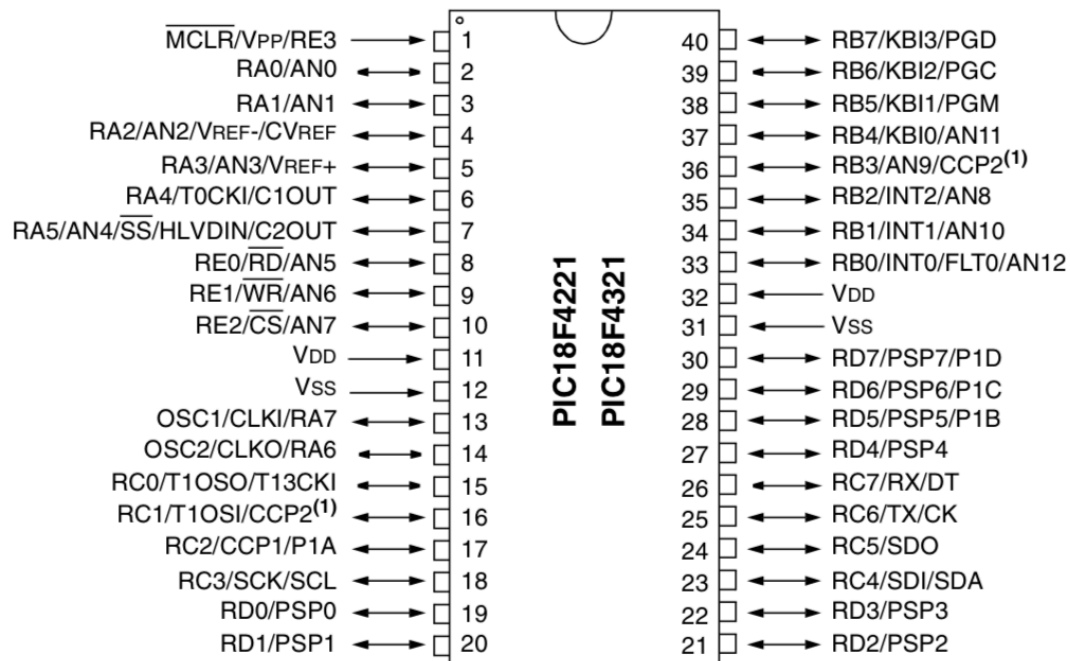
There were a few modifications to the circuit shown in image 1- VSS was actually connected to ground, VCC connected to power (5v), a 20k potentiometer was used to adjust the contrast/ brightness of the displayed characters, and there was a modification with the switches. The Pic was just directly connected to power or ground, instead of using a switch to connect it to power or ground. This was done because the DIP

switches were inefficient and would not fit in the breadboard, and because really there was no logical need for the switches. Instead, port C bit 0, 1, 2, and 3 were manually connected to either power (5v) or ground. Additionally, an input of ground was needed to turn on the display. If a switch was used, an input of ground would be achieved by closing the circuit. Additionally, my particular Oprex DMC 16249 LCD display had 14 pins instead of the typical 16 pins, as there was no back light. The pins struggled to stay in, and so to combat this issue the display had to be held by hand- in future applications, the wires could be soldered in.

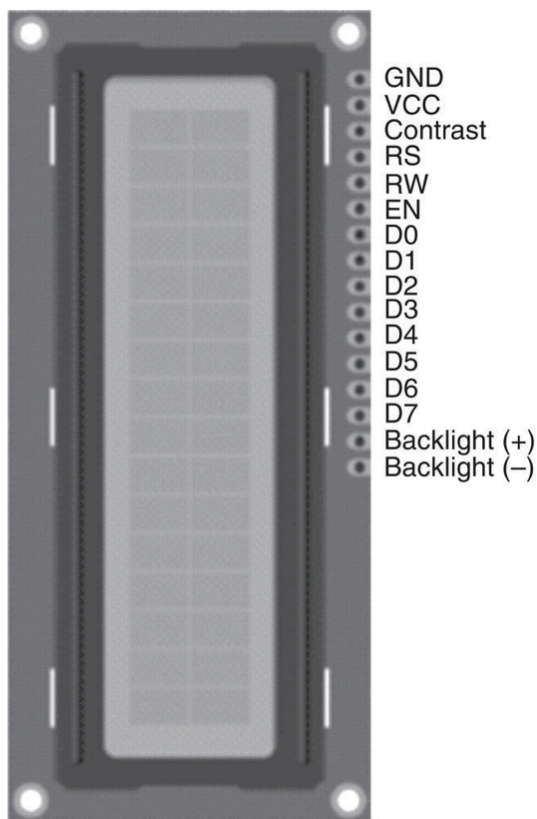


IMG 2. *The physical circuit*

To assemble the circuit, the following pinout diagrams of the Pic18F4321 and DMC 16249 was used.



IMG 3. Layout of the PIC18F4321, from microchip.com. More information is at the end of this report.



IMG 4. Layout of the Oprex DMC 16249 LCD display, from *Microcontroller Theory and Applications with the PIC18F*.

Equipment, Software, and Components required:

-Microchip's MPLAB assembler/ Debugger

Part list

- PICKit3 and PIC18F4321 chip from Microchip
- Push button for reset, DIP switches, and 20K Ω potentiometer
- Breadboard
- Resistors
- Power Supply
- Optrex DMC 16249 LCD display from Jameco.com or other suppliers
- Wires and Clip lead

Description (corresponding topics covered in the textbook)

Four DIP switches will be connected to Port C, and the Optrex DMC 16249 LCD is connected to Port D. A PIC18F assembly language program will be written to display the phrase "PIC18F" when all switches are HIGH (Section 8.4 and Section 10.10).

Prerequisites

Section 8.4

Procedure

Assemble the PIC18F assembly language programs using the MPLAB.

-Download the assembled program into the PIC18F4321 on the breadboard from your Personal Computer or Laptop using the PICKit3 and MPLAB following the steps provided in Appendix H of the book.

-Use the default clock of the PIC18F4321 and connect the appropriate RESET circuit to the PIC18F4321 pin.

-Connect the hardware to the PIC18F4321, and demonstrate the lab as a PIC18F4321-based stand-alone system.

Code

The assembly program controls an LCD display using input switches (or by directly connecting the wires to power and ground) to bits 0-3 of port C. The text 'PIC18F' is displayed when all four switches are closed, or all four inputs are 0. When any switch or input is not 0, the LCD screen is cleared.

Basic layout:

1. Define manual reset - when the reset button is pressed or the circuit is powered on, the processor jumps to the main section (main program). Also, we tell the program to put the program instructions into flash program memory (class= CODE)
2. Define the main code section and declare a variable to test if the PIC18F is currently being displayed or not. Make this variable at 0x70, and in flash memory.

MAIN

3. Set all port d bits as outputs (to LCD)
4. Set RB0-RB2 as outputs (to LCD)
5. Set RC0-RC7 as inputs (for switches)
6. Set RC4-RC7 as outputs and clear
7. Call subroutine to initialize the LCD
8. Main loop to wait for switch input- load current state of RC0-RC3 into WREG, mask upper bits, if all switches are pressed (active low) result = 0. If result is not equal to 0, go to switches off subroutine. If the switches are pressed and we have an open circuit, or input of ground, call switches on subroutine, and the display subroutine will be called. Otherwise, do nothing and continue the loop.

SWITCHES ARE ON SUBROUTINE

9. If the switches are all on, the switches on subroutine is called. If there is nothing displayed yet (variable to test if there is anything displayed on the LCD is 0 is 0) then the display text subroutine is called. Otherwise, we go back to the main loop

DISPLAY TEXT SUBROUTINE

10. Call the subroutine to clear the LCD
11. Set cursor to line 1 position 0
12. Call subroutine to send command to LCD
13. Print each character, move each character into WREG, call lcd display character subroutine, then call delay subroutine
14. Set the variable that tests if there is anything displayed on the LCD to 1, because now 'PIC18F' is displayed.

SWITCHES ARE OFF SUBROUTINE

15. If the switches are off, open and therefore not giving an output of ground, the LCD is cleared, and the variable that tests if there is anything displayed on the LCD is cleared. Then we go back to the main loop.

INITIALIZE THE LCD SETTINGS SUBROUTINE

16. Wait a bit before starting, to give LCD time to power up
17. Initialize 8 bit interface, 2 lines, and 5x8 dot characters, then send the command to the LCD
18. Display on, cursor off, no blinking, send commands to LCD
19. Initialize entry mode (increment cursor after every character and also have no screen shift), and send command to LCD
20. Clear the display and send command to LCD

SEND COMMAND TO LCD SUBROUTINE

21. Switch to bank to access port B
22. Send command
23. Get into write mode
24. Put command (in WREG) into port D (LCD data lines)
25. Make a short pulse on enable to tell LCD to accept command
26. Call delay to wait for command to finish

SEND A CHARACTER TO THE LCD SUBROUTINE

27. Access port B
28. Tell LCD to display what we send
29. Set to write
30. Display ASCII character
31. Rising edge to trigger LCD read
32. Falling edge to trigger LCD to write
33. Delay for writing the character

LONG DELAY SUBROUTINE

34. Load WREG with 200
35. Decrease by 1
36. If not 0, decrease again
37. Return from delay subroutine

*note- each instruction takes roughly $1\mu\text{s}$, one iteration is about 2 cycles, so 200 iterations is about $400\mu\text{s}$

SHORT DELAY SUBROUTINE

38. Load WREG with 20
39. Decrease by 1
40. If not 0, decrease again
41. Return from delay subroutine

*note- each instruction takes roughly $1\mu\text{s}$, one iteration is about 2 cycles, so 20 iterations is about $40\mu\text{s}$

POSTLAB:

postlab i)

-Write a C program to accomplish the Prelab

WORKS CITED

- *Microchip*, ww1.microchip.com/downloads/en/DeviceDoc/39689b.pdf. Accessed July 2025.
- Rafiquzzaman, M. "Chapter 9: PIC INTERRUPT I/O, LCD, AND KEYBOARD INTERFACING." *Microcontroller Theory and Applications with the PIC18F*, John Wiley & Sons, Inc., Canada and Hoboken, New Jersey., 2025. 9.3 PIC18F Interface to a Typical LCD (Liquid Crystal Display). Figure 9.11